FEASIBILITY REPORT

Prepared by:
Isma Haseemdeen



Web-based Platform- Fresh Grocers

➢ Background:

Fresh Grocers is a long-standing grocery delivery service in Sri Lanka, operating successfully for over fifteen years. The company has built a strong customer base by offering reliable and timely delivery of groceries to households in urban and suburban areas. Over the years, Fresh Grocers has gained a reputation for quality service, delivering a wide range of fresh produce and essential goods directly to customers' homes, contributing to its recognition as a trusted name in local grocery delivery.

The business model of Fresh Grocers traditionally relies on manual order processing and phone-based communication. Customers place orders via telephone, which are then entered and managed by customer service representatives. Delivery agents are assigned to orders based on their availability and proximity to the delivery location, often determined manually. Although effective in the past, this system lacks the flexibility, speed, and user experience demanded by today's digitally connected consumers.

To remain competitive and meet growing customer expectations, Fresh Grocers has decided to implement a modern web-based platform. This new system will enable customers to register and place orders online, receive automated SMS and email updates, select nearby delivery agents, and rate their delivery experience. It will also empower delivery agents to manage their availability and location within the system, and allow customer service staff to assist customers who are unable to place orders themselves. By embracing digital transformation, Fresh Grocers aims to streamline operations, improve customer satisfaction, and strengthen its position in the rapidly evolving online grocery market in Sri Lanka.

➢ Problem Definition

Currently, Fresh Grocers operates through a manual, telephone-based ordering system. Customers call a customer service representative to place their grocery orders, which are then recorded manually. Delivery agents are contacted individually and assigned based on their availability and approximate location, usually estimated without any real-time tracking or location data. All communications, including order confirmations and delivery updates, are handled manually through phone calls or SMS sent by staff. This process, though functional, is time-consuming and heavily dependent on human coordination. The existing system presents several critical problems. First, it lacks automation and digital tracking, which leads to delays and inefficiencies in order processing and delivery assignment. Without real-time location data, assigning the nearest delivery agent is often inaccurate, resulting in longer delivery times and higher fuel costs. Second, there is no platform for customers to view available products, track orders, or provide feedback, which limits customer engagement and satisfaction. Additionally, the current system does not support customer registration or digital authentication, making it difficult to personalize service or maintain customer order history. Furthermore, the manual handling of orders and delivery coordination increases the risk of human error, such as missed orders, incorrect deliveries, or delayed communication. The absence of automated SMS/email notifications and lack of integration between customers, delivery agents, and the customer service team reduce overall service efficiency. In an increasingly competitive market, continuing to rely on outdated processes may lead to customer attrition and lost market share. These challenges highlight the urgent need for a web-based system that can streamline operations, reduce errors, and deliver a modern, user-friendly experience.

➢ Problem Solution

To overcome the limitations of the current manual system, Fresh Grocers proposes the development and implementation of a web-based grocery ordering and delivery management platform. This system will introduce a centralized, automated solution that enhances efficiency, accuracy, and customer satisfaction. The platform will streamline the entire process; from customer registration and order placement to delivery tracking and feedback collection, by integrating modern digital technologies. The new system will allow both customers and delivery personnel to register through the website, upon which they will receive a unique username and password via email for secure login. Customers will be able to browse products, place orders online, and specify their delivery location. Upon order confirmation, they will receive an automated SMS containing the delivery agent's contact details and estimated delivery time. This automation reduces the need for manual communication and improves the speed and clarity of order handling. Additionally, the system will feature a location-based assignment module that identifies and displays the delivery agents closest to the customer's location, allowing either the customer or the customer service representative to select the most suitable agent. Delivery agents can manually update their current location through the system, while customer service staff can assist customers who prefer to place orders by phone, ensuring inclusivity. The platform will also include a customer feedback feature, enabling users to rate delivery agents after each order, which helps maintain service quality. Overall, this solution addresses the inefficiencies of the old system by digitizing key operations and delivering a faster, more transparent, and customer-friendly service experience.

➢ Scope of the System

The proposed web-based system for Fresh Grocers will focus on automating key aspects of the grocery ordering and delivery process to enhance operational efficiency and improve customer experience. The system will serve multiple user roles, including customers, delivery personnel, and customer service representatives, offering a user-friendly interface tailored to each role's needs.

The core functions that will be automated include customer and delivery agent registration, order placement, delivery assignment, location tracking, and notification services. Upon registration, users will automatically receive their login credentials via email. Customers will be able to place orders online, eliminating the need for manual order entry. Once an order is confirmed, the system will automatically send an SMS to the customer with the delivery agent's contact information and estimated delivery time.

Additionally, the system will automate the process of identifying and displaying nearby delivery agents based on the customer's location, improving the speed and accuracy of agent assignment. Delivery agents can update their location manually, and customer service representatives can input customer and location data when handling phone-based orders. The system will also automatically record customer feedback and delivery ratings after each completed order. By automating these processes, the platform minimizes human error, reduces operational delays, and ensures a more consistent and efficient service experience for all users.

➢ Objective of the System

- **Online Registration for Customers and Delivery Personnel:**
  Allow both customers and delivery agents to register on the web platform. After registration, the system will automatically send an email containing their unique username and password for secure access.

- **Web-Based Grocery Ordering:**
  Enable customers to conveniently place grocery orders through the website, streamlining the ordering process without needing phone calls or manual entry.

- **Automated SMS Notifications:**
  After an order is confirmed, the system will automatically send an SMS to the customer with the delivery agent's contact information and the estimated delivery time to keep customers informed.

- **Delivery Personnel Rating System:**
  Provide customers the ability to rate delivery agents based on their delivery experience, allowing the company to monitor service quality and improve customer satisfaction.

- **Order Entry by Customer Service Representatives:**
  Allow customer service staff to manually enter orders on behalf of customers who may not use the online system, ensuring accessibility and inclusiveness.

- **SMS Confirmation for Phone Orders:**
  When the customer service representative places an order, the system will send an SMS confirmation to the customer's phone number to confirm the order details.

- **Location-Based Delivery Agent Assignment:**

Implement a system that identifies delivery agents closest to the customer's location and displays a list of available agents. Customers or customer service representatives can then select the preferred delivery agent.

- **Delivery Agent Location Updates:**
  Enable delivery personnel to manually update their current location in the system or provide this information to customer service representatives for input, facilitating accurate delivery assignments.

- **Customer Location Specification:**
  Allow customers to specify their delivery location directly during order placement or through customer service representatives, ensuring accurate delivery routing.

➢ Stakeholders and their Roles

- **Client Company (Fresh Grocers Management):** The primary stakeholder responsible for defining business requirements, providing funding, and approving project deliverables. They oversee the project to ensure alignment with business goals and market demands.

- **Customers:** The main users of the system who place grocery orders online. Their satisfaction and ease of use are critical to the system's success. They provide feedback on system usability and delivery service quality.

- **Delivery Personnel:** Delivery agents who register on the system, update their location, and manage deliveries. Their interaction with the platform impacts the efficiency of order fulfilment.

- **Customer Service Representatives:** Staff members who assist customers by manually placing orders in the system and managing phone-based transactions. They act as intermediaries for customers who may not use the online platform.

- **Project Team:** The group responsible for planning, designing, developing, testing, and deploying the system. This includes:
    - ✓ ***System Developers and Software Engineers:*** Responsible for coding, implementing system features, and integrating functionalities as per requirements.
    - ✓ ***Quality Assurance (QA) Engineers:*** Handle system testing, identify bugs, and ensure the software meets quality standards before release.
    - ✓ ***UI/UX Designers:*** Design user-friendly interfaces ensuring the system is easy to navigate for all user groups.

- **IT Personnel (Internal Support Team):** Handle infrastructure setup, network configuration, and ongoing technical support post-deployment to maintain system stability and security.

- **Software Vendors/Third-party Providers:** External providers supplying software tools, SMS gateways, or cloud services required for functionalities such as automated notifications and hosting.

- **Finance Team:** Oversee the project budget, cost management, and financial approvals ensuring that resources are allocated efficiently.

- **Regulatory Authorities:** Ensure that the system complies with local data protection laws, electronic transaction regulations, and any other relevant standards.

➢ Feasibility Criteria

1. Operational Feasibility

- **Readiness of Existing Staff:**
  Fresh Grocers already has a structured team of customer service representatives and delivery personnel. These staff members are familiar with the current order and delivery workflows and will require only basic training to use the new system. For example, customer service representatives will learn how to manually enter customer orders through the system, while delivery agents will learn how to update their location and manage delivery assignments digitally. This ensures a smooth transition with minimal resistance or disruption.

- **Customer Willingness and Capability:**
  A significant portion of Fresh Grocers' customer base resides in city and suburban areas where internet usage and smartphone penetration are high. These customers are likely to welcome a more convenient way to order groceries online. The system also retains support for non-digital customers, ensuring no one is excluded, orders can still be placed via phone with assistance from a customer service representative, maintaining accessibility for all demographics.

- **Improved Efficiency and Service Quality:**
  Currently, many operations such as order taking, delivery coordination, and customer notifications are handled manually, which increases the risk of delays and errors. The new system automates these processes, ensuring faster order processing, accurate delivery assignments based on location, and real-time SMS updates. This will enhance customer experience and reduce the workload on staff, allowing them to focus on resolving issues and improving service quality.

- **Inclusivity and Flexibility:**
  One of the system's strengths is its flexibility. It accommodates both online and offline users. While customers who are comfortable with digital technology can use the web

platform, others can still place orders via phone. This inclusive approach ensures operational continuity for all customer segments during and after implementation.

- **Ongoing Maintenance and Support:**
  The system will be maintained by in-house IT personnel or third-party developers. This includes regular updates, bug fixes, and monitoring system performance. The organization's IT support capacity ensures that the platform will remain reliable and secure over time.

  In summary, the proposed system is highly compatible with Fresh Grocers' existing operational model. It enhances key areas such as order processing, delivery coordination, customer interaction, and staff productivity. With proper training and support, the system can be successfully integrated and operated within the organization, resulting in improved service delivery and long-term operational efficiency.

2. Technical Feasibility

- **Hardware Requirements**
  a) **Server Machine:** A dedicated server will be required to host the database and application backend. This server will manage all system processes, including order handling, user management, and communications with third-party services like SMS and map APIs. It should have sufficient RAM (at least 8–16GB), a multi-core processor, and SSD storage to ensure fast data retrieval and processing. The server must also have backup capabilities and uninterrupted power supply (UPS) support for reliability.
  b) **Branded Desktop PCs:** Medium-range branded desktop computers (e.g., Dell or HP) will be allocated for the use of customer service representatives and administrative staff. These machines will be used for tasks such as manually placing orders, accessing customer records, assigning delivery personnel, and generating reports. These brands offer durability, technical support, and compatibility with the proposed software stack, which reduces hardware-related risks.

c) **Printing Equipment:** An inkjet printer will be included in the setup for generating physical records such as receipts, order confirmations, or delivery summaries. While most operations are digital, printing capabilities ensure flexibility and documentation when required.

- **Network Infrastructure**
  a) **Local Area Network (LAN):** The office setup will require a secure LAN to connect the server with the desktop PCs. This setup allows real-time sharing of data such as order status, inventory changes, and customer queries between systems, ensuring efficient internal communication.
  b) **Internet Connectivity:** A stable and high-speed internet connection is necessary for enabling access to the web-based system by customers and delivery personnel. Internet access also allows the system to send SMS notifications and retrieve location data using APIs. It is essential that the connection is reliable to avoid service interruptions.
  c) **Networking Devices:** Basic networking hardware like routers, switches, and Ethernet cables will be used to establish connectivity between the server and client machines. These devices form the backbone of communication and ensure data is transferred efficiently and securely.

- **Software Requirements**
  a) **Development Environment:** The system will be developed using the C# programming language within Microsoft Visual Studio 2017 IDE. This environment supports rapid development, debugging, and integration with other Microsoft tools. It is highly suitable for enterprise-grade applications and well-documented, making maintenance easier.
  b) **Database Management System:** Microsoft SQL Server 2017 will be used to store user data, order history, delivery records, and other transactional information. It is a reliable and secure relational database system that ensures data integrity and supports concurrent transactions.
  c) **SMS and Location APIs:** The system will integrate with SMS gateway services (such as Twilio or a local provider) to send automatic notifications to customers after order placement. Additionally, integration with Google Maps API will

allow real-time proximity calculation between customers and delivery agents, enabling efficient agent selection.

d) **Web Interface Framework:** The web front-end may be developed using ASP.NET MVC or a similar framework that supports a responsive, clean, and user-friendly interface for all stakeholders, including customers, delivery personnel, and administrative staff.

- **Technical Support**
  a. **Experienced Development Team:** The project team has previous experience using the proposed tools (C#, SQL Server, Visual Studio). This minimizes technical learning curves and ensures that common challenges can be resolved quickly based on prior knowledge.
  b. **Availability of Local Resources:** All hardware and software required for the project are easily available in Sri Lanka. This means that there will be no delays in sourcing equipment or licenses, and local vendors can provide quick replacements or upgrades if needed.
  c. **Maintenance and Future Upgrades:** The system will be designed with scalability and maintainability in mind. Regular updates, bug fixes, and feature enhancements can be managed either by in-house IT personnel or outsourced software developers. Proper documentation and version control practices will be followed for smooth transitions.
  d. **User Training and Documentation:** To ensure ease of use, end users such as customer service agents and delivery personnel will be trained on how to use the system. Training sessions, user manuals, and quick-start guides will be provided. This guarantees effective adoption and reduces dependency on technical staff for daily operations.

3. Economic Feasibility

INITIAL COST	
Development cost	200,000
Hardware cost	350,000
Software License cost	50,000
Implementation and Deployment cost	30,000
Subscription fee (third-party services)	20,000
Total Initial cost: 650,000	

ONGOING COSTS (Every year)	
Electricity and IT infrastructure	100,000
Maintenance and Technical Support	120,000
Software Subscription renewal	30,000
Internet and Communication costs	40,000
System upgrades and enhancements	50,000
Security and Backup services	20,000
Total Initial cost: 360,000	

EXPECTED BENEFITS	
Increased Revenue	400,000
Cost Savings and Staff Reduction	200,000
Total Initial cost: 600,000	

Year	Cost (LKR)	Benefit (LKR)	Net Benefit (LKR)	Accumulated Net Benefit (LKR)
0	650,000	0	0	0
1	360,000	600,000	240,000	240,000
2	360,000	600,000	240,000	480,000
3	360,000	600,000	240,000	720,000
4	360,000	600,000	240,000	960,000

Payback Period

Payback Period= Initial Investment/ Net Annual Benefit = 650,000/240,000

≈ 2.7 years

This indicates the project recovers its costs in about **2 years and 7 months**, which is reasonable for such an investment.

Return on Investment (ROI)

Using Year 4 accumulated net benefit:

ROI = (Net Income / Cost of Investment) × 100 = (960,000 / 2090, 000) × 100 ≈ 45.93%

An ROI of approximately **46%** signals a strong financial return and supports moving forward with the project.

4. Schedule Feasibility

The client, Fresh Grocers, expects the web-based grocery delivery system to be fully developed and operational within **two months**. The development is scheduled to begin in the first week of July 2025, and the system is expected to go live by early September 2025**.** To determine if this timeframe is realistic, an effort estimation has been performed based on the required components of the system such as data entry forms and reports.

Effort Estimation

Component	Number Required	Lines of Code per Item	Total Lines of Code
Data Entry Forms	12 (Registration, Login, Order Placement, Location Entry, Order Assignment, etc.)	300	300 × 12 = 3,600
Reports	5 (Delivery Report, Customer Feedback, Daily Orders, Agent Ratings, Revenue Report)	100	100 × 5 = 500
			Total: 4,100 Lines of Codes

Time Allocation Chart

Phase	Estimated Time	Feasibility	Description
Requirements Analysis	5 working days	**High**	Stakeholder requirements are clear and well-defined, making this phase straightforward.
System Design	5 working days	**High**	Includes ER diagrams, UI wireframes, and workflow planning. Scope is small to medium scale, manageable within timeframe.
Coding & Unit Testing	7 working days	**Medium**	Requires implementation of approximately 4,100 lines of code. With two developers, this is realistic, but some complexity may arise.
User Acceptance Testing	2 working days	**High**	Since only key features will be tested by end-users, this short phase is feasible.

User Manual Preparation	4 working days	**High**	Standard documentation for basic user and admin functions. Time allotted is sufficient.
Buffer (Contingencies)	3 working days	**High**	Built-in buffer helps address unexpected issues such as bugs, feature tweaks, or delays.

- Total Time Required: 26 working days (Approximately 5.2 weeks)
- Time Available: 2 months (8 weeks)
- Considering a 5-day work week, some public holidays, and occasional sick leave, the project can realistically be completed in under 1.5 months. Since the client has allowed 2 full months, this schedule provides enough flexibility to manage risks and delays.
- **Conclusion:** With all phases rated mostly high in feasibility and a reasonable buffer included, the project is schedule feasible.

5. Legal Feasibility

Legal feasibility assesses whether the proposed system complies with relevant laws, regulations, and industry standards. For Fresh Grocers, the web-based grocery ordering and delivery system is considered legally feasible, based on the following considerations:

1. Data Protection and Privacy Laws: The system will handle sensitive customer data such as names, phone numbers, locations, and order details. Therefore, the platform must comply with Sri Lanka's Data Protection Act No. 9 of 2022, which mandates: Explicit user consent for data collection and usage, secure storage and processing of personal data, and allowing users to access, correct, or delete their data. Proper privacy policies and user agreements will be included in the system to ensure legal compliance.

2. Electronic Transactions: As the system supports online ordering and communication via email and SMS, it must comply with the Electronic Transactions Act No. 19 of 2006. This includes: Recognizing electronic communications as legally valid, ensuring transaction integrity and authenticity, and providing secure channels for communication between users and the system.

3. Employment and Labor Regulations: The platform involves registration and assignment of delivery personnel. Therefore, Fresh Grocers must ensure compliance with: Employment laws, including fair working conditions, working hours, and payment regulations for delivery agents, Proper contracts or service agreements must be in place if delivery agents are third-party contractors.

4. Consumer Rights and Fair Trading: The system must adhere to consumer protection laws such as: Providing accurate information on groceries, prices, and delivery times, ensuring fair billing practices and refund/cancellation policies, and enabling customer complaints and rating mechanisms.

5. Software Licensing: All development tools and software used (e.g., C#, MS SQL Server, Visual Studio) must be properly licensed. The use of open-source libraries must comply with their respective licenses (e.g., MIT, GPL).

Conclusion: The proposed system is legally feasible as long as it incorporates appropriate measures to protect personal data, ensure secure online transactions, follow employment standards, respect consumer rights, and use properly licensed software tools. Legal consultation may be done during development to ensure full compliance.

➢ Risk Assessment for the System

Risk	Description	Impact Level	Likelihood	Mitigation Strategy
Data Breach or Unauthorized Access	Sensitive customer and delivery data may be accessed by hackers or unauthorized users.	High	Medium	Use encryption, secure authentication, role-based access control, and conduct regular security audits.
System Downtime or Server Failure	Server or internet failures could render the system unusable.	High	Medium	Use reliable hosting, backup servers, regular maintenance, and monitoring systems.
Inaccurate Delivery Assignments	Outdated or incorrect location data may assign the wrong delivery agent, delaying delivery.	Medium	Medium	Ensure frequent location updates from agents and validate user input.
SMS and Email Delivery Failures	SMS or email messages with login details and confirmations might not be delivered.	Medium	High	Use trusted APIs, add retry mechanisms, and offer alternative in-platform notifications.
User Resistance to Adoption	Some users may be reluctant or find it difficult to use the new system.	Medium	Medium	Provide tutorials, a user-friendly design, and ongoing customer support.
Project Timeline Overruns	The system may not be completed by the scheduled deadline	High	Medium	Follow a clear project schedule, allocate buffer

	due to unforeseen delays.			days, and track progress regularly.
Cost Overruns	The project may go over budget due to unexpected feature requests or technical issues.	**Medium**	**Medium**	Monitor budget closely, define scope clearly, and prioritize essential features.
Legal or Compliance Issues	Failure to comply with data privacy and digital transaction laws may result in penalties.	**High**	**Low**	Consult legal advisors and ensure compliance with all applicable regulations during development.

While risks such as data breaches, system unavailability, and deadline for the project overruns carry considerable effect, the possibility of these situations can be decreased with robust security protocols, reliable infrastructure, and smart project management. Medium-level risks, including improper delivery assignments, communication failures, and user resistance, can be mitigated by assuring timely updates, employing reliable communication services, and developing an intuitive user interface with enough support. Although legal compliance concerns exist, they are rather low in likelihood and may be addressed by consulting legal specialists throughout development. Considering the thorough identification of these risks and realistic solutions, the project team is well-prepared to tackle potential hurdles. This balanced understanding and preparedness imply that the system is suitable for development. The predicted benefits of streamlining operations, enhancing customer experience, and improving delivery coordination outweigh the controllable risks, making the proposed system a practical and strategically useful investment for Fresh Grocers.

➢ Alternative Technical Solutions with Cost/Benefit Analysis

Given the outlined cost/benefit analysis, several alternative technical solutions can be considered to improve cost-effectiveness while maintaining or enhancing system performance. One option is to utilize open-source technologies instead of licensed software. For instance, replacing paid tools like Microsoft Visual Studio and SQL Server with open-source alternatives such as Visual Studio Code and PostgreSQL could significantly reduce software licensing costs. This shift would lower the initial software cost and eliminate annual subscription fees, thereby improving the system's ROI and shortening the payback period.

Another potential solution is adopting a cloud-based software-as-a-service (SaaS) model instead of building a fully customized system from scratch. This approach could reduce development and maintenance costs since cloud platforms often offer built-in security, scalability, and infrastructure management. It may also eliminate the need for in-house servers and reduce electricity and IT infrastructure costs. Cloud platforms like Amazon Web Services (AWS), Microsoft Azure, or Google Cloud could provide scalable options with pay-as-you-go pricing models that align with the growth of Fresh Grocers.

Additionally, outsourcing certain components of the development or technical support to third-party vendors could reduce internal staffing costs and speed up deployment. Hybrid app development using cross-platform frameworks such as Flutter or React Native could also reduce development time and cost if mobile app support is later considered.

These alternatives, when evaluated against the existing economic feasibility data, can potentially reduce the initial and ongoing expenses, increase the net benefit, and deliver a faster payback period with an even stronger return on investment.

➢ Recommendations

Based on the comprehensive feasibility study, it is strongly recommended that Fresh Grocers proceed with the development and implementation of the proposed web-based ordering and delivery system. To ensure success, the company should adopt a development approach that balances reliability and cost-effectiveness, potentially exploring both proprietary and open-source technologies to optimize long-term expenses. Utilizing cloud-based infrastructure is also advisable, as it can reduce upfront hardware costs, enhance scalability, and improve data security and system maintenance. The system should incorporate role-based access control to protect sensitive information and ensure proper user permissions for customers, delivery agents, and staff. Key operational tasks such as customer registration, order placement, delivery agent assignment, and SMS notifications should be automated to improve efficiency and reduce manual errors. Additionally, a strong plan for ongoing maintenance and technical support must be established to maintain system stability and address issues promptly. Comprehensive training programs for staff and delivery personnel will be essential for smooth adoption and effective use of the system. Once deployed, continuous monitoring of performance indicators and user feedback should be implemented to guide ongoing improvements. Finally, the system architecture should be designed with future scalability in mind, allowing Fresh Grocers to incorporate additional features such as mobile app support and online payment options as the business grows. Following these recommendations will help ensure the system delivers maximum value and supports the company's long-term growth and competitive advantage.

The End